

Lab Presentation



Undergraduate Research
Opportunity Program (UROP)



UNIVERSITÉ
**PARIS
SCIENCES
& LETTRES**

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Surface Instability of Yield-Stress Fluids

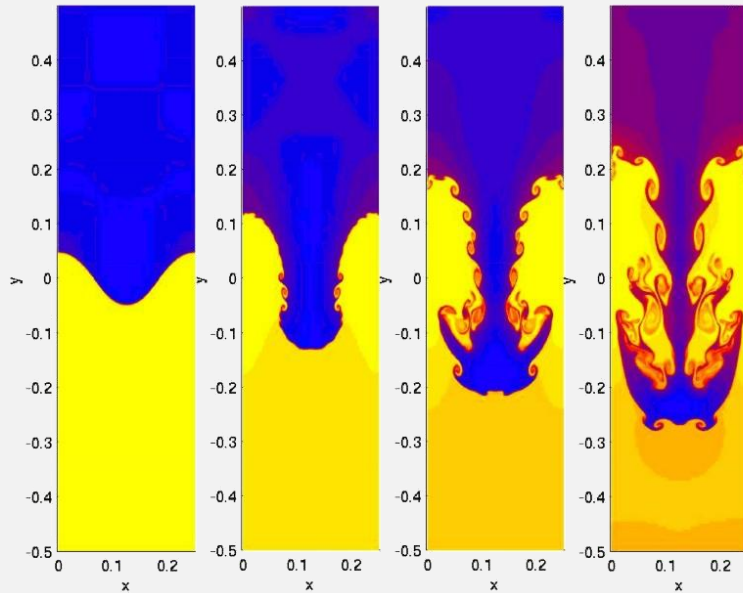
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Soft Matter Science & Engineering team
ESPCI – PSL, Paris, France

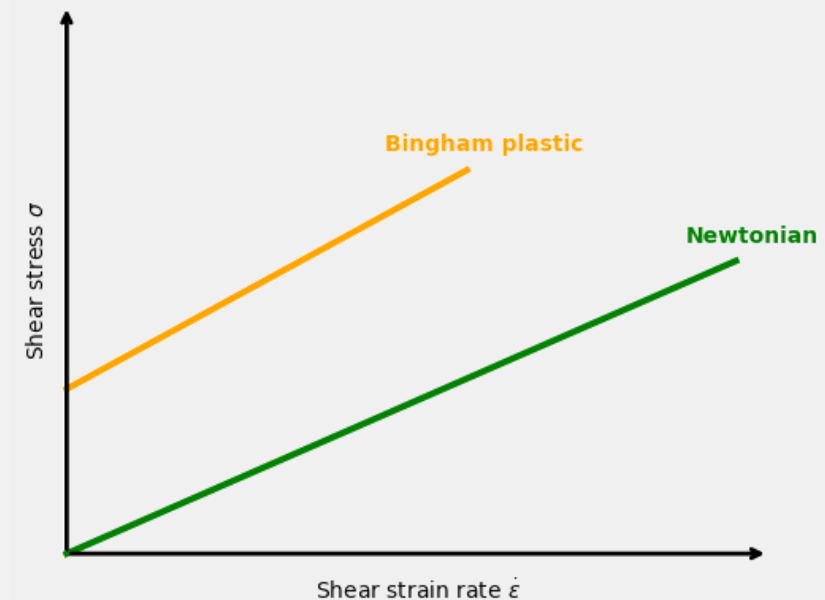
Rayleigh-Taylor Instability (RTI) and Yield-Stress Fluids



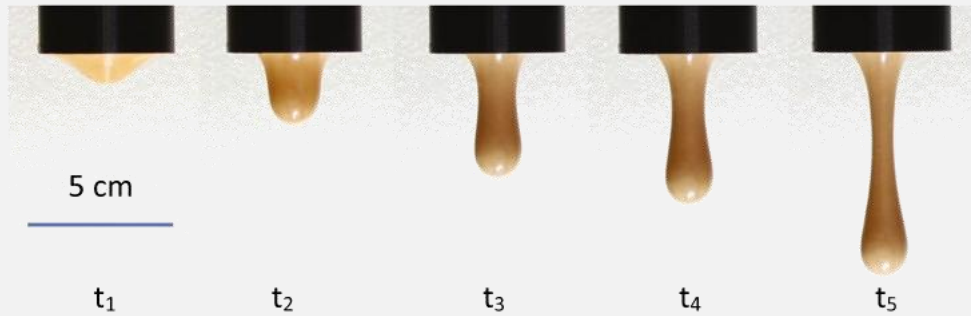
Simulation of an RTI "finger"
from Los Alamos National Laboratory

Stress/Shear-rate graph for Newtonian
and Yield-stress fluids

Shear stress as a function of the shear strain rate

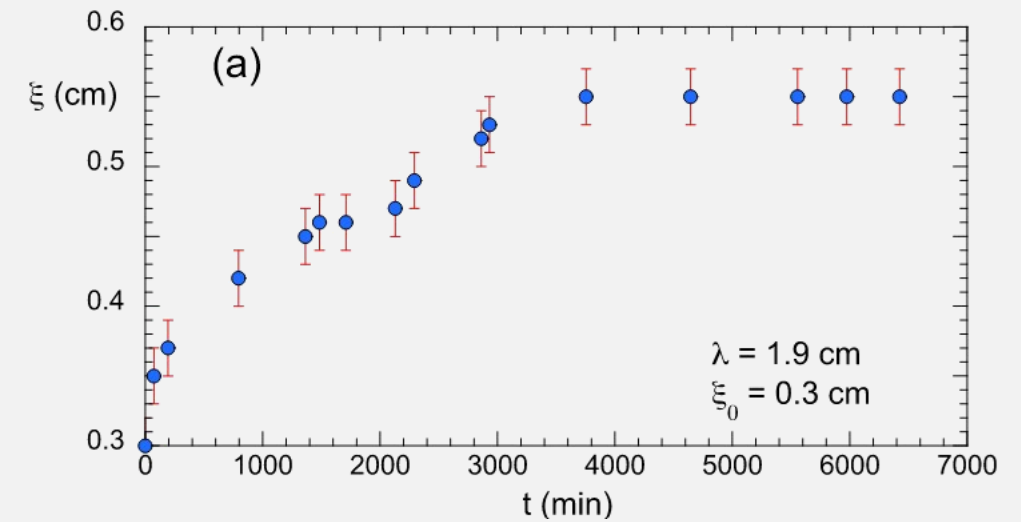


Piriz et al. experiment and results (2025)



Evolution snapshots of a typical unstable case

Instantaneous perturbation amplitude ξ as a function of time for a stable case





Objectives of the internship and table of contents

Objectives

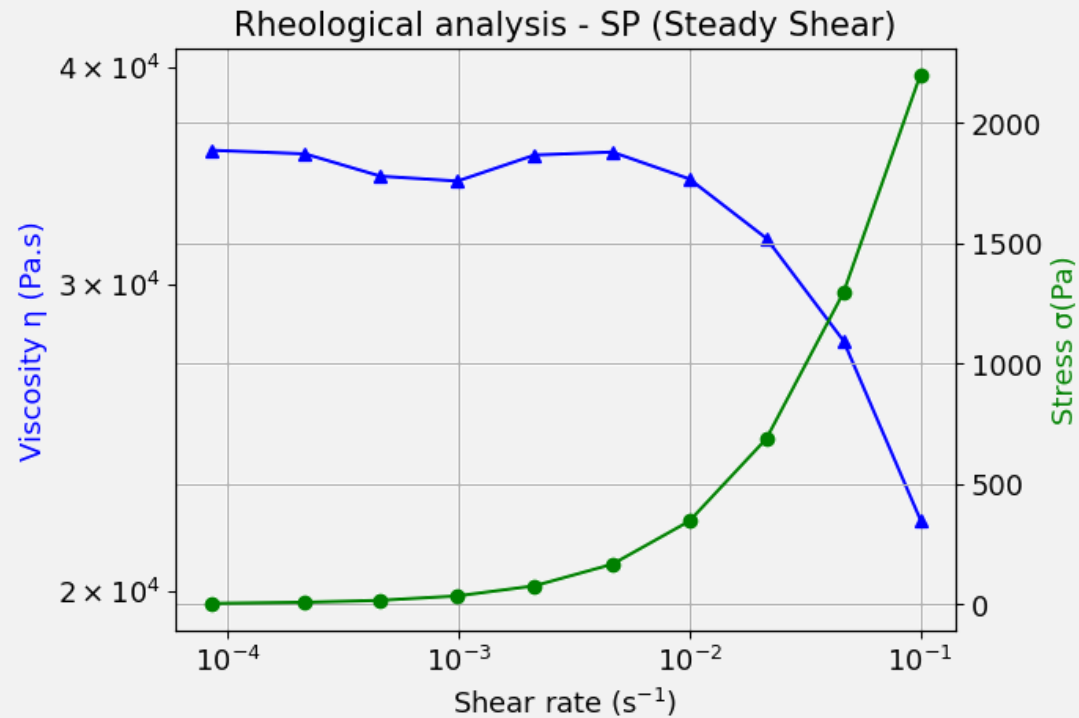
- Determine the yield-stress fluid behavior of **Silly Putty (SP)**
- Develop a possible explanation for Piriz' results

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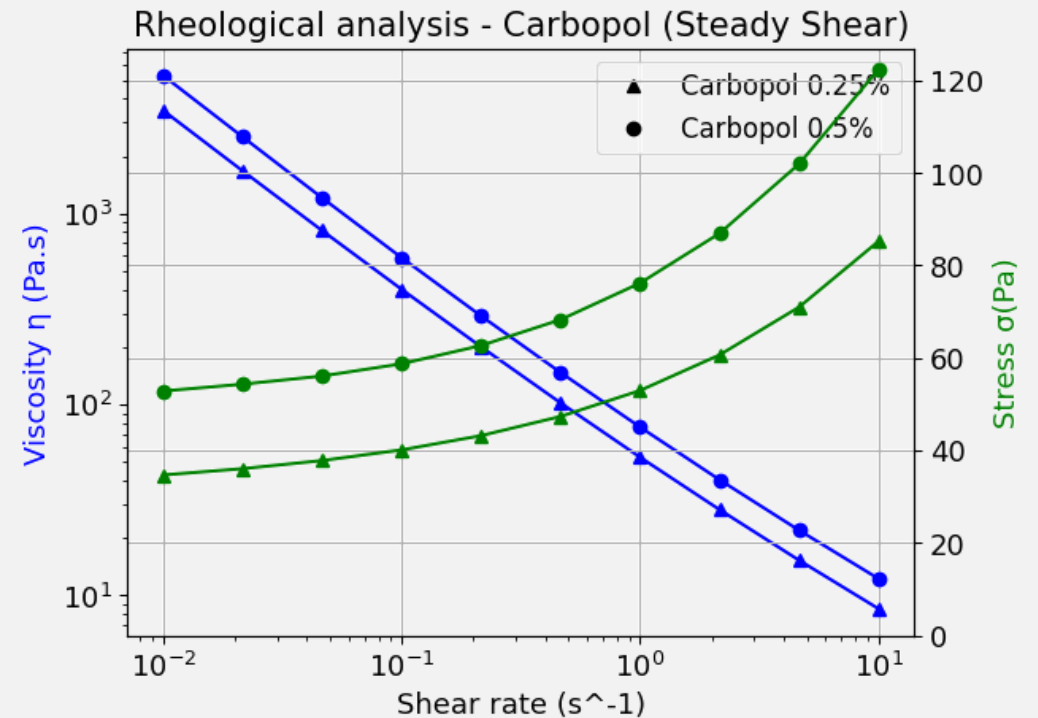
1. Rheological study of Silly Putty
2. Relaxation experiment with laser-sheet
3. Combined Instability – Relaxation experiment with laser-sheet



Rheological study of SP and comparison with Carbopol



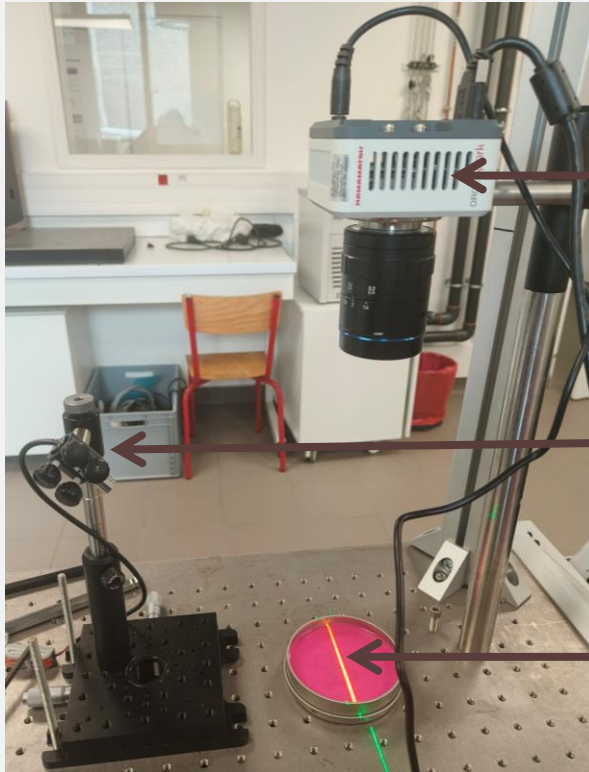
Steady Shear on Silly Putty



Steady Shear on a yield-stress fluid

Conclusion → Silly Putty does not exhibit a yield-stress

First Laser-sheet Experiment

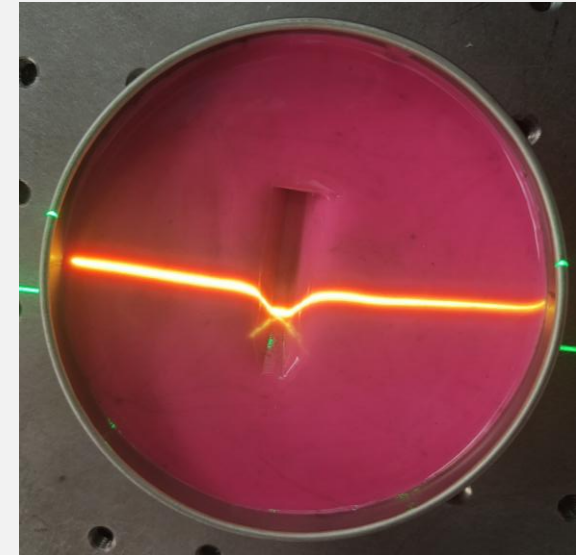


Camera

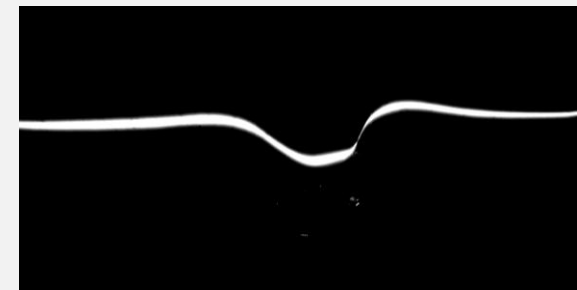
Laser-sheet

SP sample

Experimental set-up

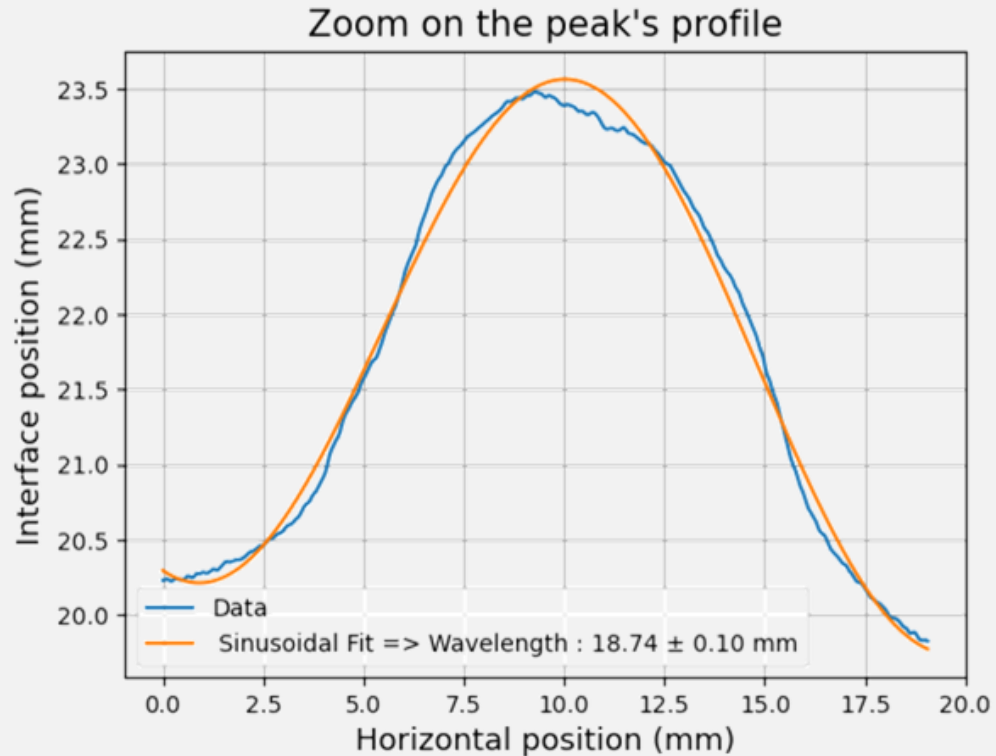


Laser-line on the SP sample

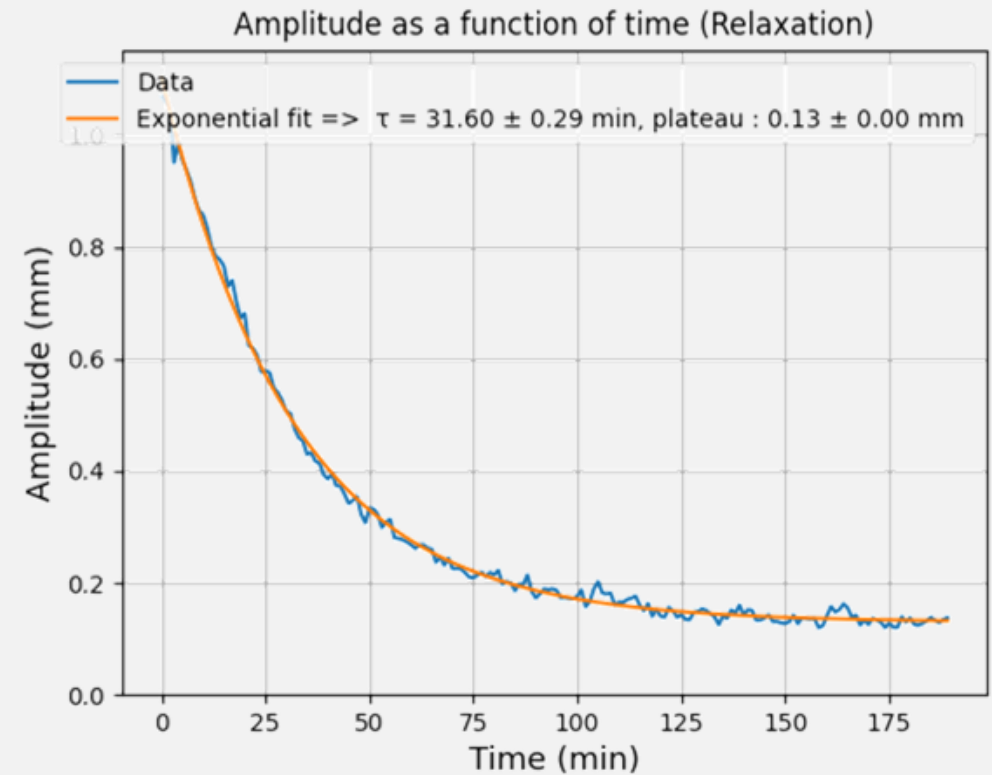


Camera snapshot of the laser-line

First Laser-sheet experiment - Results



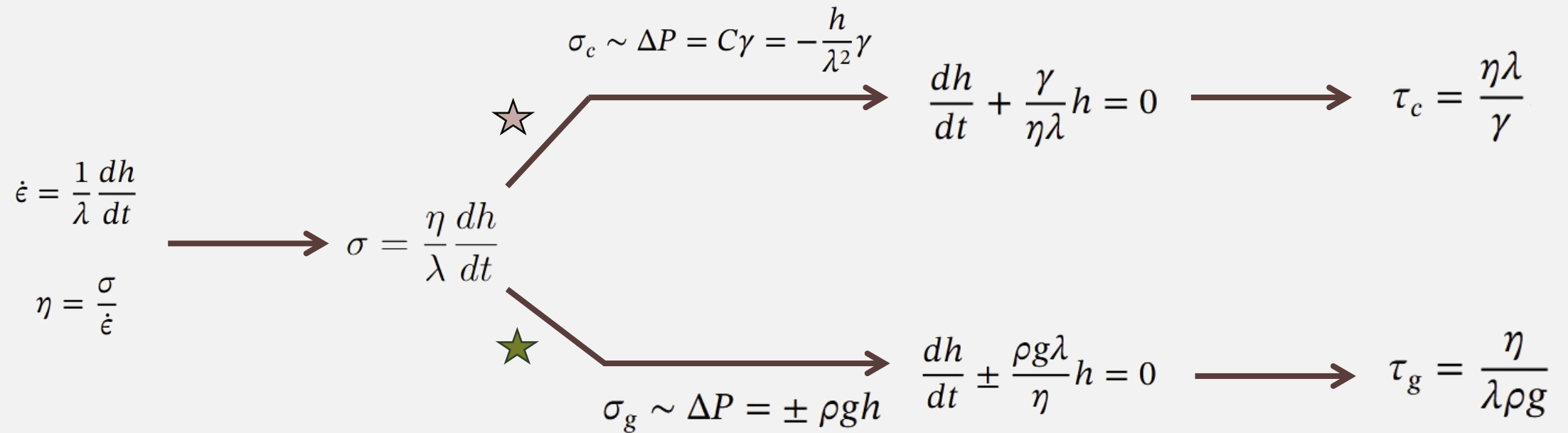
Extraction of a wavelength λ
with a sinusoidal fit of the profile



Extraction of a characteristic time τ
with an exponential fit of the amplitude



Theoretical characteristic times

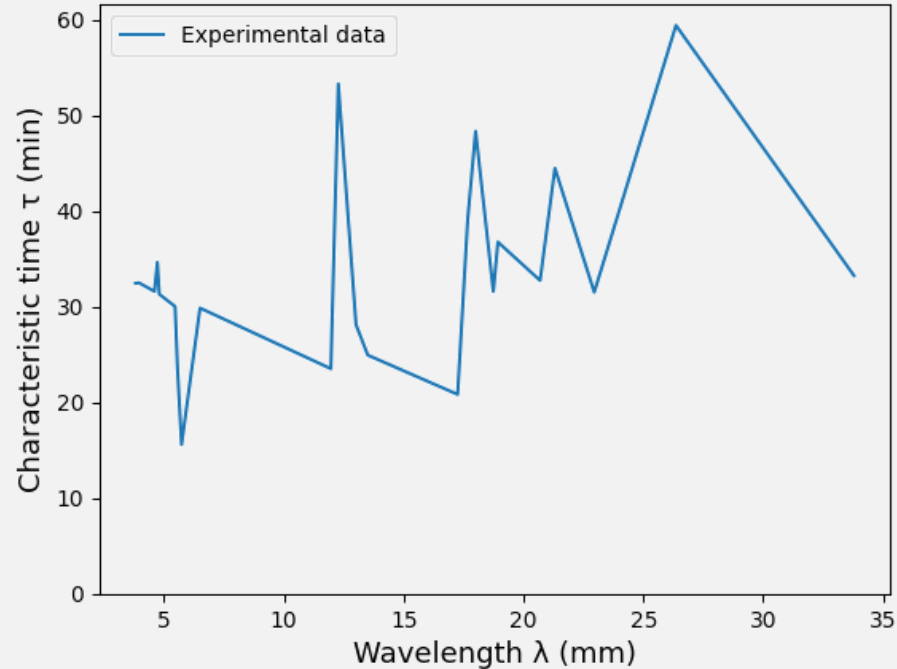


★ Surface-tension-driven

★ Gravity-driven

Dependance between characteristic time τ and wavelength λ

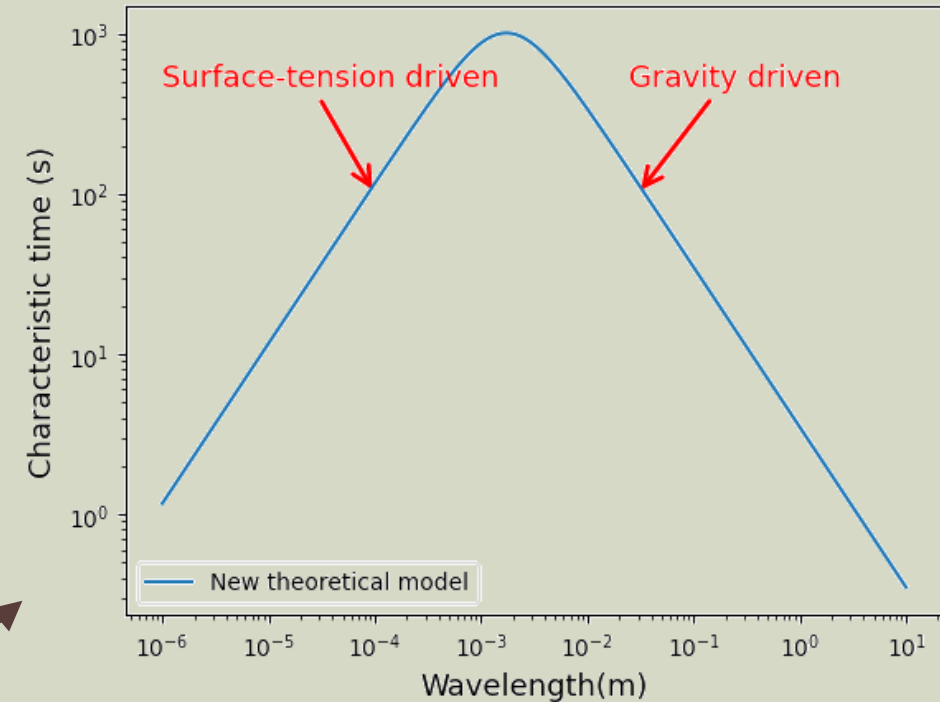
Evolution of the characteristic time as a function of the wavelength



New hypothesis :

$$\frac{1}{\tau} = \frac{1}{\tau_g} + \frac{1}{\tau_c} \longrightarrow \tau(\lambda) = \frac{1}{\frac{\gamma}{\eta\lambda} + \frac{\lambda\rho g}{\eta}}$$

Characteristic time as a function of the wavelength



Analytical solution for the transition :

$$\lambda_{transition} = l_c = \sqrt{\frac{\gamma}{\rho g}}$$

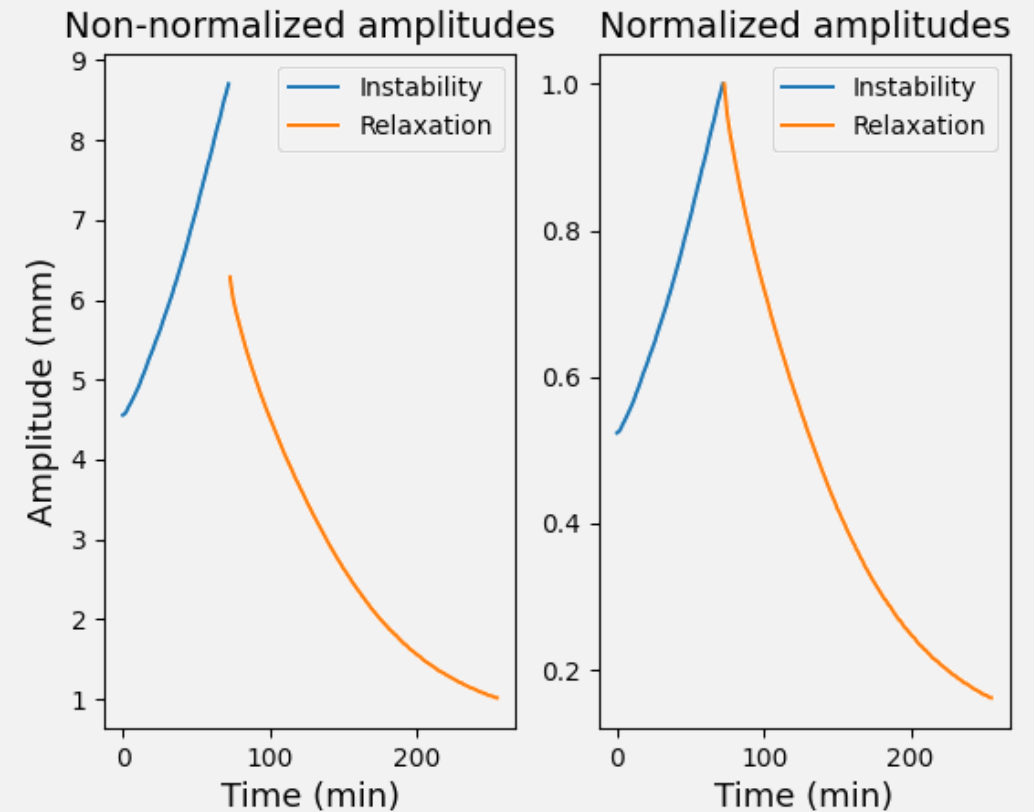
Second Laser-sheet experiment



SP sample

Laser-sheet

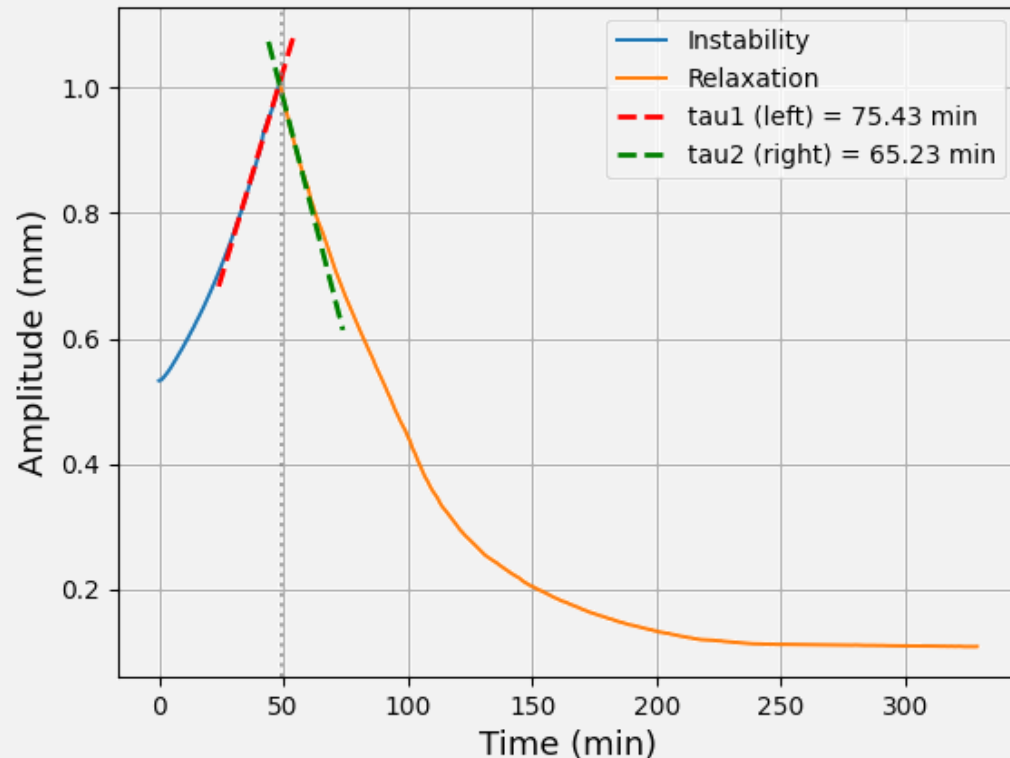
Camera



Comparison before and after normalization

Second Laser-sheet experiment - Results

Evolution of the amplitude over time



Characteristic times :

- Instability : $\frac{1}{\tau_1} = \frac{1}{\tau_g} - \frac{1}{\tau_c}$
- Relaxation : $\frac{1}{\tau_2} = \frac{1}{\tau_g} + \frac{1}{\tau_c}$



Extraction of a characteristic time for each regime



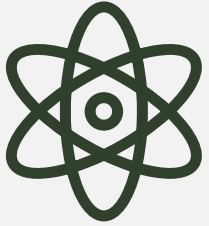
Characteristic times for different experiments

	Instability (τ_1)	Relaxation (τ_2)	Surface-tension-driven (τ_c)	Gravity-driven (τ_g)
Exp1	65,08	34,72	148,9	45,28
Exp2	73,27	38,52	162,4	50,49
Exp3	81,04	58,17	412,3	67,73
Exp4	87,30	67,07	578,9	75,86



Take-Home Messages

- **The non-Yield-stress fluid behavior of Silly Putty**
- **The significant role of surface-tension in the stabilization observed by Piriz *et. al* 2025**



Thank you for your attention

Any question ?

